

Data Documentation (Real-World)

I found out it's not a good idea to use entire countries to represent simple, idealized pattern like even spreads, perfect clusters, or smooth gradual gradients for real-world mapping because most countries are huge and contain possible types of distribution.

The core purpose of my research is to compare map performance under specific known conditions. A large country hybrid/complex distribution would simply act as a single, messy scenario, making it hard to isolate why one mapping technique failed and succeeded. In this case, subnational regions would be more suitable for this research.

Here are the identified regions;

Three regions from three different countries (Poland, USA and Germany) were considered for this study.

Evenly Distributed - Łódzkie Voivodeship

Łódzkie Voivodeship is a province in Poland that contains low to moderate density, agricultural background that approximates a uniform spread.

The entire Poland population data was captured and i narrowed down to focus on Łódzkie Voivodeship Province.



Projection: Lambert Azimuthal Equal Area (ETRS 1989 LAEA)

Population data

Source: WorldPop <https://hub.worldpop.org/geodata/summary?id=74975>

Administrative Boundaries

Source: GADM https://geodata.ucdavis.edu/gadm/gadm4.1/shp/gadm41_POL_shp.zip

Highly Clustered - Los Angeles County, USA

This region is a classic example of an extreme, sharp cluster. Most of the population is packed tightly into the narrow coastal plain, directly adjacent to large, uninhabited mountain ranges and deserts.

The entire USA population data was captured and i narrowed down to focus on Los Angeles County.



Projection: Lambert Conformal Conic (NAD 1983 StatePlane California V FIPS 0405 (US Feet))

Population data

Source: WorldPop: <https://hub.worldpop.org/geodata/summary?id=75903>

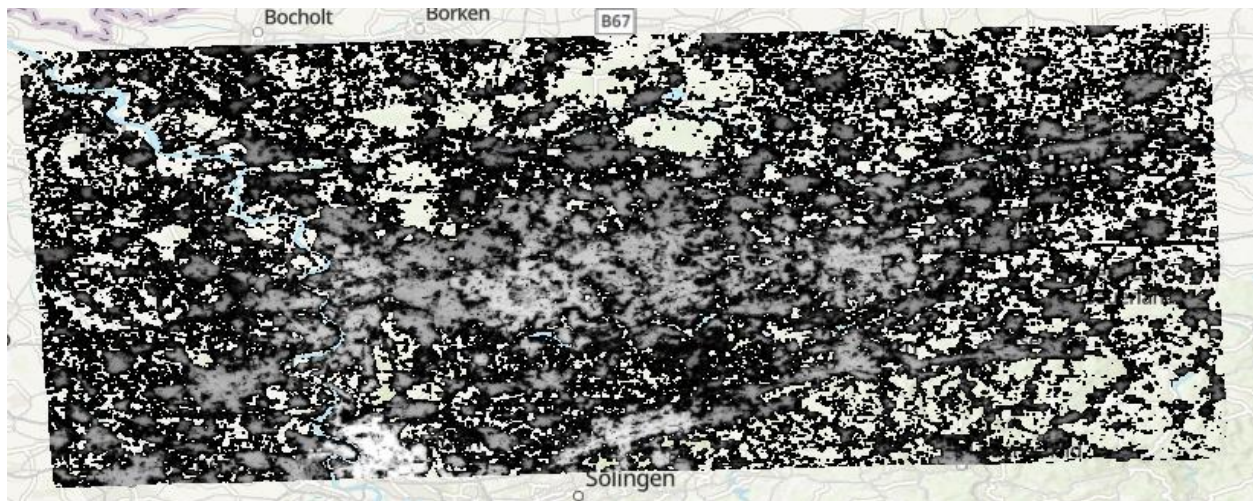
Administrative Boundaries

Source: GADM: https://geodata.ucdavis.edu/gadm/gadm4.1/shp/gadm41_USA_shp.zip

Ruhr Region, Germany – Gradual Gradient

The Ruhr region is a well-developed, polycentric urban area where many cities have grown together into a single urban mass. This merging creates a smooth, continuous density field where the population gradually fades toward rural areas.

The entire Germany population data was captured and i selected the following cities; Essen, Dortmund, Bochum, Hagen, Hamm, Wesel, Unna, Bottrop, Herne, Gelsenkirchen, Oberhausen, Ennepe-Ruhr-Kreis, Duisburg, Mulheim an der Ruhr, Recklinghausen to make up the Ruhr region.



Projection: Lambert Azimuthal Equal Area (ETRS 1989 LAEA)

Population data

Source: <https://hub.worldpop.org/geodata/summary?id=73535>

Administrative Boundaries

Source: https://geodata.ucdavis.edu/gadm/gadm4.1/shp/gadm41_DEU_shp.zip

Tools: ArcGIS Pro

Reference

1. Bondarenko M., Priyatikanto R., Tejedor-Garavito N., Zhang W., McKeen T., Cunningham A., Woods T., Hilton J., Cihan D., Nosatiuk B., Brinkhoff T., Tatem A., Sorichetta A.. 2025 Constrained estimates of 2015-2030 total number of people per grid

square at a resolution of 3 arc (approximately 100m at the equator) R2025A version v1. Global Demographic Data Project - Funded by The Bill and Melinda Gates Foundation (INV-045237). WorldPop - School of Geography and Environmental Science, University of Southampton. DOI:10.5258/SOTON/WP00839